

CIND 820: Big Data Analytics Project

Architecture and Data Audit

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Introduction

Title: Insurance Fraud Detection Using Machine Learning on Claims Data

Insurance fraud remains one of the most significant challenges facing the global insurance industry. Fraudulent claims result in substantial financial losses for insurers, increased premiums for policyholders, and reduced operational efficiency. The Coalition Against Insurance Fraud estimates that insurance fraud costs billions of dollars annually across various insurance sectors [1]. As insurance transactions continue to become increasingly digitized, traditional fraud detection methods are proving insufficient for identifying sophisticated fraudulent activities. Consequently, researchers and practitioners have increasingly adopted data analytics and machine learning techniques to improve fraud detection capabilities.

The application of data analytics in insurance fraud detection has evolved considerably over the past two decades. Early approaches primarily relied on rule-based systems and statistical methods, while contemporary research emphasizes machine learning algorithms capable of identifying complex patterns within large volumes of insurance claims data. However, despite advances in predictive modeling, concerns regarding model interpretability, transparency, and class imbalance remain significant challenges within the fraud detection domain [2].

This literature review examines the evolution of insurance fraud detection methodologies, including statistical techniques, machine learning approaches, explainable artificial intelligence (XAI), and current challenges associated with fraud analytics. The review concludes by identifying an appropriate methodological approach for our project.

Traditional Approaches to Insurance Fraud Detection

Historically, insurance companies relied heavily on expert-driven rule-based systems for fraud identification. These systems used predefined business rules such as unusually high claim amounts, multiple claims submitted within a short period, or inconsistencies between policy information and claim details. While rule-based systems provided transparency and ease of implementation, researchers found that they struggled to adapt to evolving fraud patterns and frequently generated high false-positive rates [3].

To address these limitations, statistical methods such as Logistic Regression became widely adopted in insurance fraud detection. Logistic Regression models the probability of fraud by estimating relationships between explanatory variables and a binary fraud outcome. Viaene et al. compared several classification techniques for automobile insurance fraud detection and found Logistic Regression to be an effective baseline due to its interpretability and ease of

implementation [4]. The primary advantage of Logistic Regression is that model coefficients provide clear explanations regarding how individual variables influence fraud probability. However, the model assumes linear relationships among variables and may fail to capture complex interactions that frequently characterize fraudulent behavior [4].

Machine Learning Approaches for Insurance Fraud Detection

Recent studies demonstrate that machine learning techniques frequently outperform traditional statistical models in fraud detection tasks. Machine learning algorithms can identify complex nonlinear relationships and interactions among variables that are difficult to detect through conventional approaches. Several supervised learning methods have emerged as dominant techniques within insurance fraud analytics. Below we will discuss the commonly used machine learning approaches in the industry along with pros and cons of each approach.

1. Decision Trees

Decision Trees are among the earliest machine learning techniques applied to fraud detection problems. These models classify observations through hierarchical decision rules based on feature values. Researchers have highlighted the interpretability of Decision Trees as a major advantage because the resulting decision paths can be easily understood by business stakeholders and fraud investigators [5].

However, Decision Trees are susceptible to overfitting, particularly when datasets contain noisy observations or severe class imbalance. Individual trees often exhibit instability, meaning small changes in the training data can produce significantly different decision structures. These limitations have motivated the development of ensemble learning techniques designed to improve predictive performance.

2. Random Forest

Random Forest extends Decision Trees by constructing multiple trees using random subsets of observations and features. Predictions are generated through aggregation of individual tree outputs. Baesens et al. reported that Random Forest models consistently outperform individual Decision Trees in fraud detection tasks because they reduce variance and improve generalization performance [6].

Random Forest algorithms are particularly effective in handling nonlinear relationships and high-dimensional datasets. Furthermore, they provide measures of feature importance that offer some degree of interpretability. Despite these advantages, Random Forest models remain less transparent than Logistic Regression because individual predictions are derived from the combined outputs of numerous decision trees [6].

3. Support Vector Machines

Support Vector Machines (SVMs) have also been widely applied to fraud detection. SVMs seek to identify an optimal decision boundary that maximizes separation between fraudulent and non-fraudulent observations. Phua et al. identified SVMs as one of the most effective classification methods for various fraud detection applications due to their ability to handle high-dimensional data and complex decision boundaries [7].

Nevertheless, SVMs present practical limitations. Model training can become computationally expensive for large datasets, and the resulting classification boundaries are often difficult to interpret. These challenges reduce their suitability in highly regulated industries such as insurance, where transparency and explainability are essential requirements [7].

4. Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) has emerged as one of the most effective machine learning algorithms for insurance fraud detection. XGBoost builds trees sequentially, allowing each new tree to correct errors made by previous trees.

Mabunda demonstrated that XGBoost achieved superior performance compared to Random Forest and other traditional machine learning approaches when applied to insurance fraud datasets [8]. The algorithm effectively handles nonlinear relationships, large feature spaces, missing values, and class imbalance. Furthermore, built-in regularization mechanisms help reduce overfitting and improve model robustness.

Despite its strong predictive capabilities, XGBoost introduces interpretability challenges. The internal decision-making process is significantly more complex than that of Logistic Regression or Decision Trees, leading many researchers to classify XGBoost as a semi-black-box model [9]. This limitation has contributed to growing interest in Explainable Artificial Intelligence techniques.

Explainable Artificial Intelligence in Insurance Fraud Detection

The increasing adoption of advanced machine learning models has raised concerns regarding transparency and accountability. Insurance organizations operate within highly regulated environments where fraud detection decisions can have significant financial and legal consequences. Consequently, stakeholders often require explanations regarding why specific claims are classified as suspicious.

Owens et al. argue that explainability is a fundamental requirement for insurance analytics because automated decisions must be understandable to investigators, regulators, and policyholders [10]. Explainable Artificial Intelligence (XAI) seeks to address this challenge by providing mechanisms for interpreting complex model behavior.

Among contemporary XAI methods, SHAP (SHapley Additive Explanations) has emerged as one of the most widely adopted techniques. SHAP provides both global and local explanations by quantifying the contribution of individual features to model predictions. Similarly, Local Interpretable Model-Agnostic Explanations (LIME) offers interpretable approximations of complex models for individual observations [9].

The literature increasingly suggests that combining high-performing machine learning algorithms with explainability frameworks provides a practical balance between predictive accuracy and transparency. Such approaches allow organizations to leverage advanced analytics while satisfying regulatory and operational requirements [10].

Class Imbalance Problem in Insurance Fraud Detection

One of the most persistent challenges in insurance fraud analytics is class imbalance. Fraudulent claims typically represent only a small proportion of total claims, resulting in datasets heavily dominated by legitimate transactions. This imbalance can cause machine learning models to achieve high overall accuracy while performing poorly on the minority fraud class.

Ngai et al. emphasize that traditional accuracy metrics can be misleading in fraud detection contexts because models may simply predict the majority class and still achieve high accuracy scores [5]. Consequently, researchers increasingly recommend evaluation metrics such as Precision, Recall, F1-Score, and Area Under the Receiver Operating Characteristic Curve (ROC-AUC).

Several approaches have been proposed to address class imbalance, including Random Oversampling, Random Under sampling, Synthetic Minority Oversampling Technique (SMOTE), and cost-sensitive learning. Recent studies indicate that combining sampling techniques with ensemble machine learning algorithms can substantially improve fraud detection performance [8]. However, no single imbalance handling strategy has been universally accepted, suggesting that techniques should be selected based on the characteristics of individual datasets.

Emerging Trends in Insurance Fraud Analytics

Recent literature highlights several emerging directions in fraud detection research.

Graph-based analytics have gained attention for their ability to identify relationships among claimants, vehicles, service providers, and organized fraud networks. Unlike traditional tabular models, graph approaches capture connections among entities and can uncover coordinated fraud schemes that would otherwise remain hidden. However, these methods introduce additional complexity and explainability challenges

Researchers are also investigating anomaly detection and unsupervised learning approaches for situations where labeled fraud examples are unavailable. These methods identify unusual patterns without requiring prior fraud labels, potentially enabling the discovery of previously unknown fraud strategies. Nevertheless, unsupervised approaches often exhibit lower precision and can generate higher false positive rates compared to supervised learning models.

Additionally, advances in generative AI have introduced new fraud risks. Fraudsters increasingly use AI-generated images, fabricated documents, and synthetic evidence to support fraudulent claims. As a result, insurers are adopting advanced analytics and AI-assisted detection systems to counter these emerging threats.

Research Gap and Methodological Justification

The reviewed literature demonstrates that advanced machine learning algorithms such as Random Forest and XGBoost frequently outperform traditional statistical methods in insurance fraud detection. However, several gaps remain evident.

First, many studies prioritize predictive performance while providing limited discussion regarding explainability and transparency. Given the regulatory requirements of the insurance industry, interpretability remains a critical consideration that cannot be overlooked [10].

Second, existing research often treats predictive accuracy and model interpretability as competing objectives. Few studies systematically evaluate whether the performance gains achieved by advanced machine learning techniques justify the reduction in transparency associated with more complex models [9].

Third, class imbalance continues to present significant methodological challenges. While numerous imbalance-handling techniques have been proposed, there remains limited consensus regarding best practices across different insurance fraud datasets [8].

In this project will adopt a comparative modeling approach. Logistic Regression will serve as the baseline model due to its transparency and interpretability. XGBoost will be implemented as an advanced alternative because of its demonstrated effectiveness in fraud detection tasks.

Model performance will be evaluated using precision, recall, F1-score, and ROC-AUC rather than accuracy alone. Additionally, feature importance analysis and explainability techniques will be incorporated. This approach provides a balance between predictive performance and interpretability while addressing the unique characteristics of insurance fraud data.

Data Profiling and Audit

Overview of the Dataset

The dataset utilized in this project was sourced from Kaggle Inc and is publicly accessible under the title " Vehicle Claim Fraud Detection Dataset " [\[11\]](#).

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15420 entries, 0 to 15419
Data columns (total 33 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Month                                15420 non-null  object
1   WeekOfMonth                          15420 non-null  int64
2   DayOfWeek                           15420 non-null  object
3   Make                                15420 non-null  object
4   AccidentArea                        15420 non-null  object
5   DayOfWeekClaimed                    15420 non-null  object
6   MonthClaimed                        15420 non-null  object
7   WeekOfMonthClaimed                  15420 non-null  int64
8   Sex                                 15420 non-null  object
9   MaritalStatus                       15420 non-null  object
10  Age                                  15420 non-null  int64
11  Fault                               15420 non-null  object
12  PolicyType                          15420 non-null  object
13  VehicleCategory                     15420 non-null  object
14  VehiclePrice                        15420 non-null  object
15  FraudFound_P                        15420 non-null  int64
16  PolicyNumber                        15420 non-null  int64
17  RepNumber                           15420 non-null  int64
18  Deductible                          15420 non-null  int64
19  DriverRating                        15420 non-null  int64
20  Days_Policy_Accident                 15420 non-null  object
21  Days_Policy_Claim                    15420 non-null  object
22  PastNumberOfClaims                  15420 non-null  object
23  AgeOfVehicle                         15420 non-null  object
24  AgeOfPolicyHolder                   15420 non-null  object
25  PoliceReportFiled                   15420 non-null  object
26  WitnessPresent                      15420 non-null  object
27  AgentType                           15420 non-null  object
28  NumberOfSupplements                 15420 non-null  object
29  AddressChange_Claim                 15420 non-null  object
30  NumberOfCars                        15420 non-null  object
31  Year                                15420 non-null  int64
32  BasePolicy                          15420 non-null  object
dtypes: int64(9), object(24)
memory usage: 3.9+ MB
```

Data Audit

The insurance fraud dataset used in this project consists of structured tabular data containing information related to insurance policies, customers, claims, vehicles, and incident characteristics. The primary objective of the dataset is to support the identification of potentially fraudulent insurance claims through exploratory data analysis and predictive modeling.

The target variable for this project is the fraud indicator, which classifies claims as either fraudulent or non-fraudulent. The dataset contains a mixture of numerical, categorical, and date-related variables that describe both the policyholder and the claim event.

Some of dataset attributes include:

- Customer demographics (age, gender, education level)
- Policy information (policy type, policy premium, policy duration)
- Vehicle information (vehicle make, vehicle age, vehicle category)
- Claim details (claim amount, incident severity, number of vehicles involved)
- Fraud label (fraud Found: Yes/No)

```
In [5]: df.describe().T
```

Out[5]:

	count	mean	std	min	25%	50%	75%	max
WeekOfMonth	15420.00	2.79	1.29	1.00	2.00	3.00	4.00	5.00
WeekOfMonthClaimed	15420.00	2.69	1.26	1.00	2.00	3.00	4.00	5.00
Age	15420.00	39.86	13.49	0.00	31.00	38.00	48.00	80.00
FraudFound_P	15420.00	0.06	0.24	0.00	0.00	0.00	0.00	1.00
PolicyNumber	15420.00	7710.50	4451.51	1.00	3855.75	7710.50	11565.25	15420.00
RepNumber	15420.00	8.48	4.60	1.00	5.00	8.00	12.00	16.00
Deductible	15420.00	407.70	43.95	300.00	400.00	400.00	400.00	700.00
DriverRating	15420.00	2.49	1.12	1.00	1.00	2.00	3.00	4.00
Year	15420.00	1994.87	0.80	1994.00	1994.00	1995.00	1996.00	1996.00

The dataset is considered structured data because each record follows a consistent schema and is stored in a tabular format suitable for statistical analysis and machine learning applications.

Exploratory Data Analysis (EDA)

Univariate Analysis

Univariate analysis examines individual variables independently to understand their distributions, identify anomalies, and evaluate overall data quality

Numerical Variables

Key numerical variables will be analyzed including:

- Age
- Policy Annual Premium
- Claim Amount
- Vehicle Age
- Number of Vehicles Involved
- Total Claim Amount

For each numerical variable, descriptive statistics will be calculated, including:

- Mean
- Median
- Minimum
- Maximum
- Standard Deviation
- Quartiles

Histograms and boxplots will be generated to visualize distributions and detect potential outliers.

Categorical Variables

Frequency distributions will be generated for all categorical features, including:

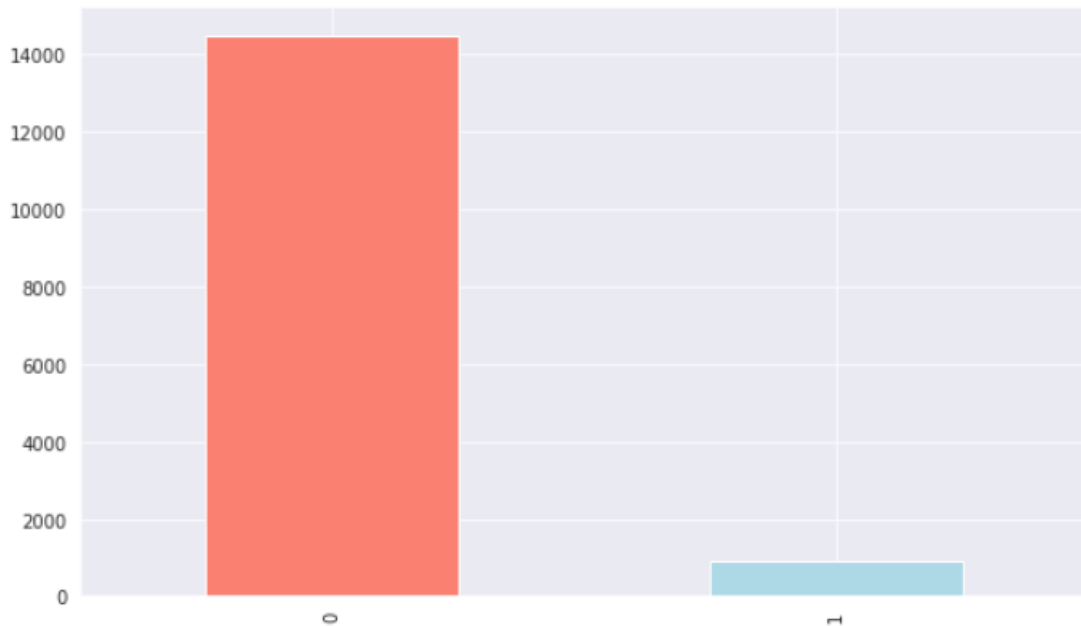
- Gender
- Vehicle Make
- Policy Type
- Vehicle Category
- Marital Status
- Accident Area
- Incident Type
- Fraud Found

Bar charts will be used to visualize category frequencies and identify dominant classes

Total Fraud vs Non-Fraud Cases in the dataset:

```
In [6]: df["FraudFound_P"].value_counts()
```

```
Out[6]:  
0    14497  
1     923  
Name: FraudFound_P, dtype: int64
```



Sex Distribution in the Dataset (Male vs Female)

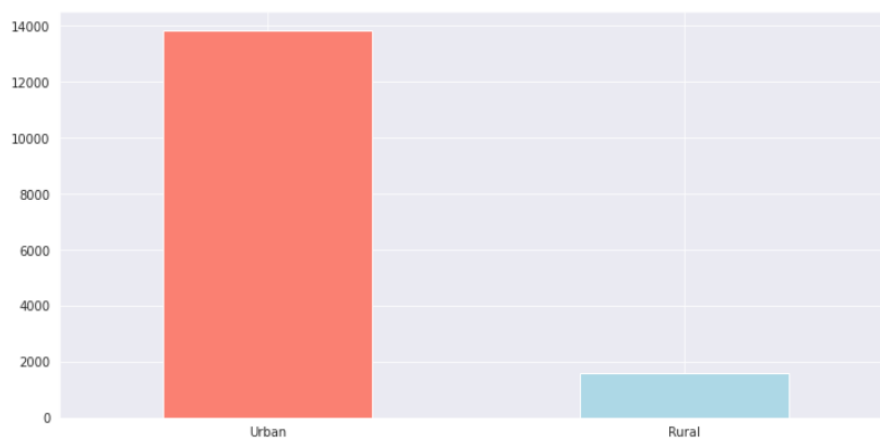
```
In [8]: df["Sex"].value_counts()
```

```
Out[8]: Male      13000  
        Female    2420  
        Name: Sex, dtype: int64
```

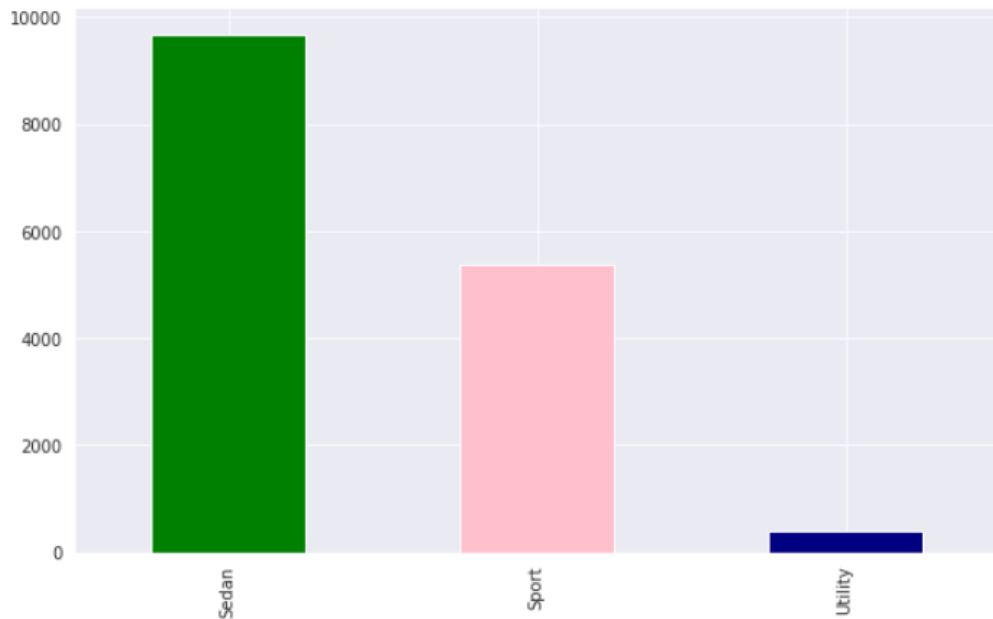
Accident Area Distribution (Urban vs Rural)

```
In [10]: df.AccidentArea.value_counts()
```

```
Out[10]: Urban      13822  
         Rural      1598  
         Name: AccidentArea, dtype: int64
```



Vehicle Category Distribution (Sedan vs Sport vs Utility)



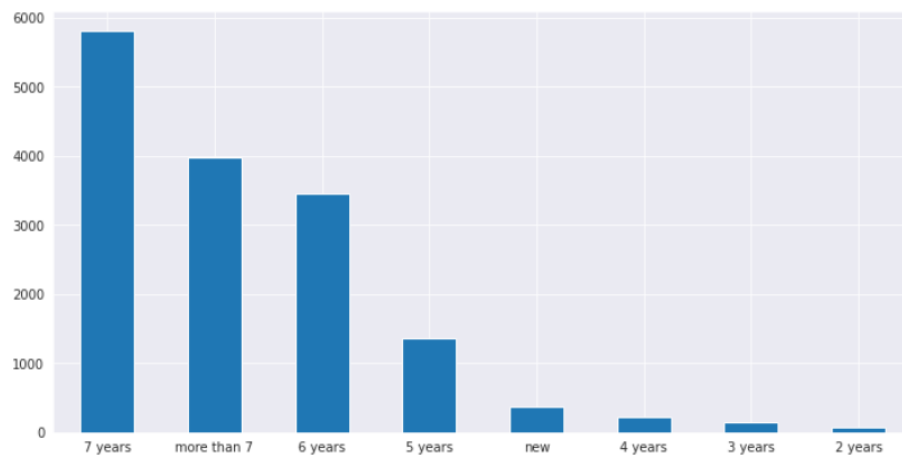
Vehicle Age Distribution

```
In [13]: df.AgeOfVehicle.value_counts()
```

```
Out[13]:
```

7 years	5807
more than 7	3981
6 years	3448
5 years	1357
new	373
4 years	229
3 years	152
2 years	73

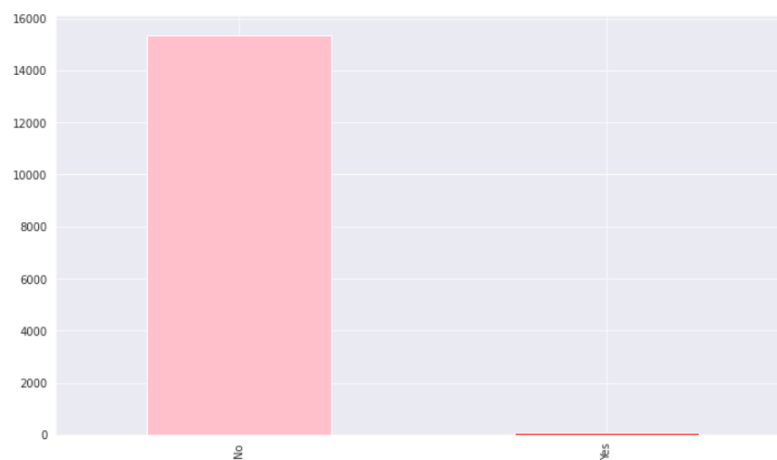
Name: AgeOfVehicle, dtype: int64



Witness Present

```
In [15]: df.WitnessPresent.value_counts()
```

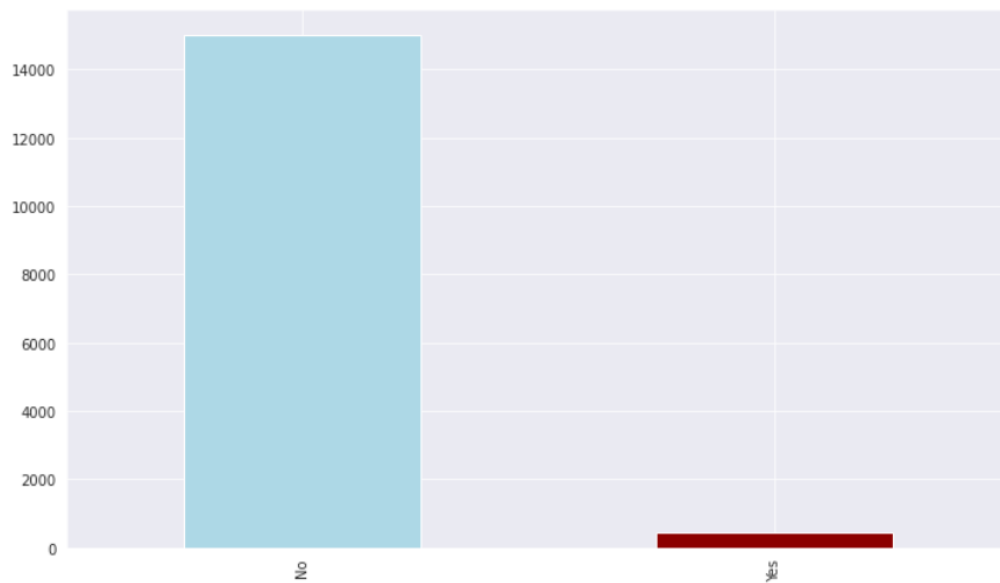
```
Out[15]: No      15333  
         Yes       87  
         Name: WitnessPresent, dtype: int64
```



Police Report Filed

```
In [17]: df.PoliceReportFiled.value_counts()
```

```
Out[17]:  
No      14992  
Yes       428  
Name: PoliceReportFiled, dtype: int64
```



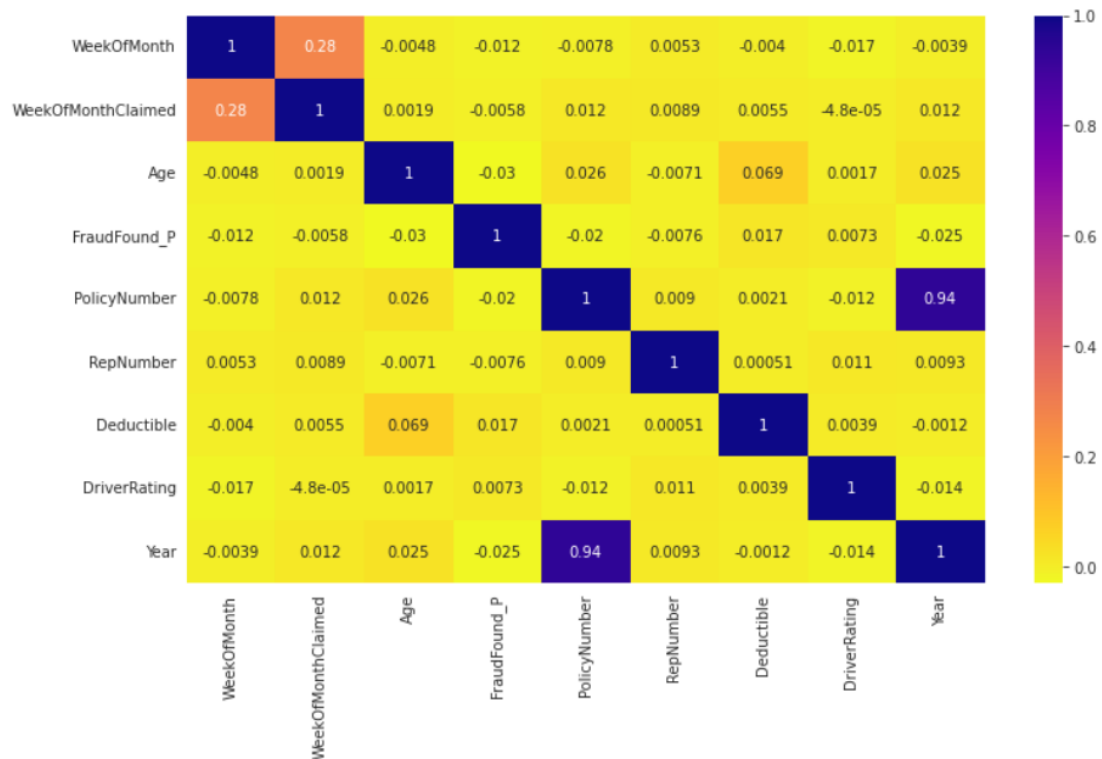
Feature Correlation

In [21]:
df.corr()

Out[21]:

	WeekOfMonth	WeekOfMonthClaimed	Age	FraudFound_P	PolicyNumber	RepNumber	Deductible	DriverRating
WeekOfMonth	1.00	0.28	-0.00	-0.01	-0.01	0.01	-0.00	-0.02
WeekOfMonthClaimed	0.28	1.00	0.00	-0.01	0.01	0.01	0.01	-0.00
Age	-0.00	0.00	1.00	-0.03	0.03	-0.01	0.07	0.00
FraudFound_P	-0.01	-0.01	-0.03	1.00	-0.02	-0.01	0.02	0.01
PolicyNumber	-0.01	0.01	0.03	-0.02	1.00	0.01	0.00	-0.01
RepNumber	0.01	0.01	-0.01	-0.01	0.01	1.00	0.00	0.01
Deductible	-0.00	0.01	0.07	0.02	0.00	0.00	1.00	0.00
DriverRating	-0.02	-0.00	0.00	0.01	-0.01	0.01	0.00	1.00
Year	-0.00	0.01	0.03	-0.02	0.94	0.01	-0.00	-0.01

Correlation between independent and dependent (target Variable)

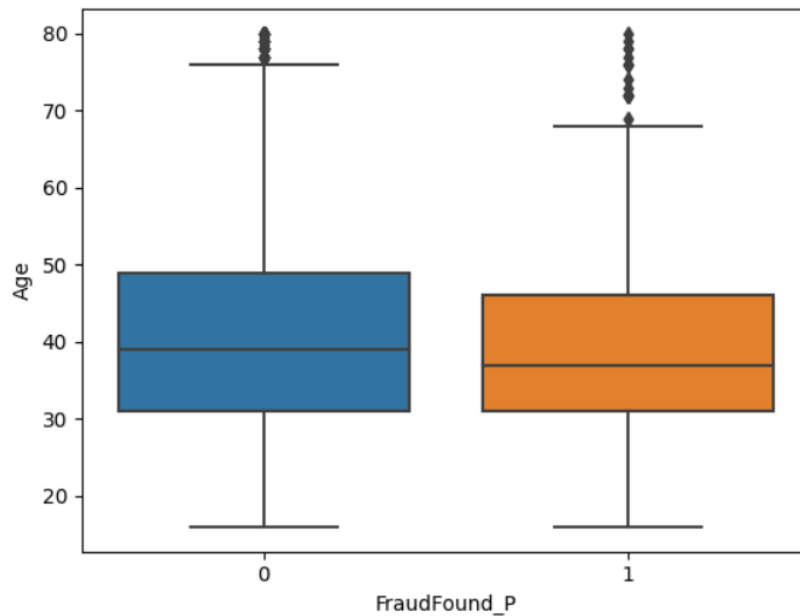


Bivariate Analysis

Bivariate analysis examines relationships between variables and identifies factors potentially associated with fraudulent behavior.

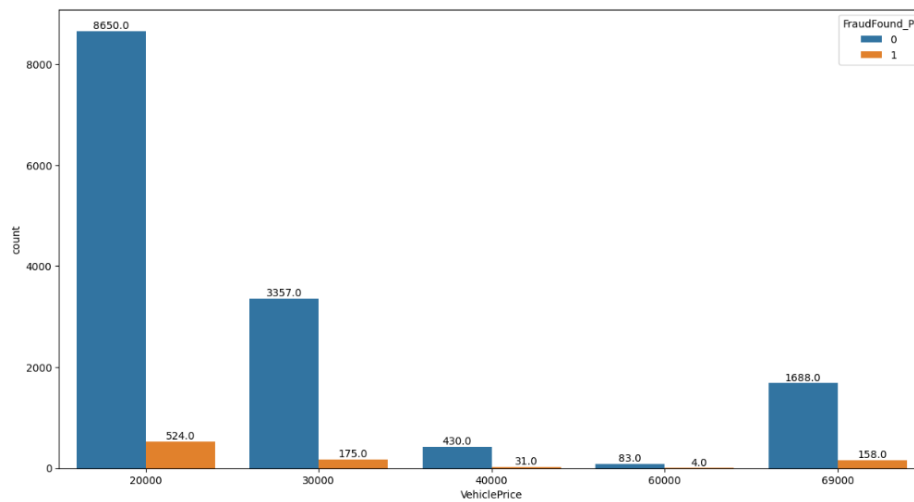
Average Age vs Fraud Found

A comparison of average age across fraudulent and non-fraudulent claims will be performed using boxplots and summary statistics.



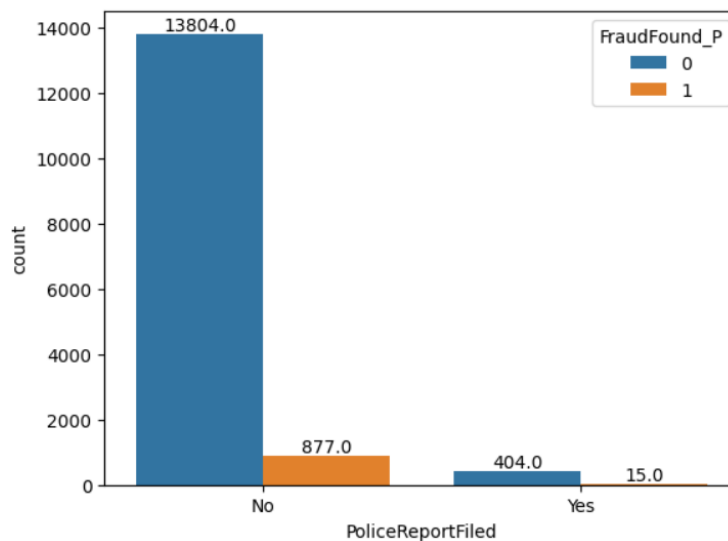
Price of Vehicle vs Fraud Found

A comparison of price of vehicle across fraudulent and non-fraudulent claims will be performed using boxplots and summary statistics to explore the association between the price of vehicles and the likelihood of fraud. The proportions of fraudulent claims differ across different price categories of vehicles.



Presence of Police Report vs Fraud Found

Association between the presence of a police report and the occurrence of fraud will be explored. The proportions of fraudulent claims can differ between claims with and without a police report.

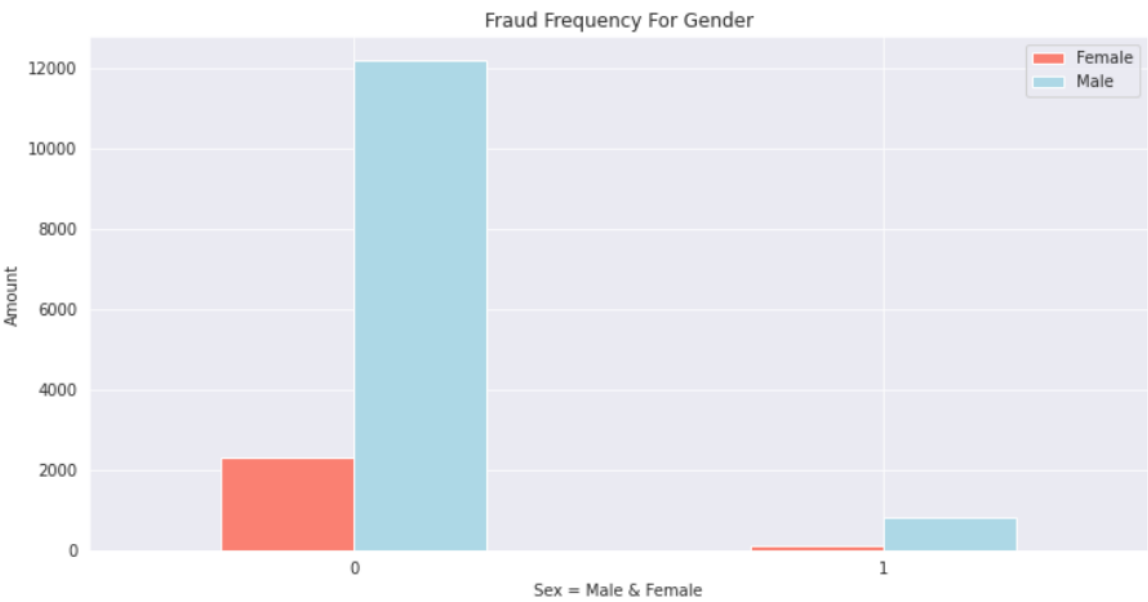


Type of Policy vs Fraud Found

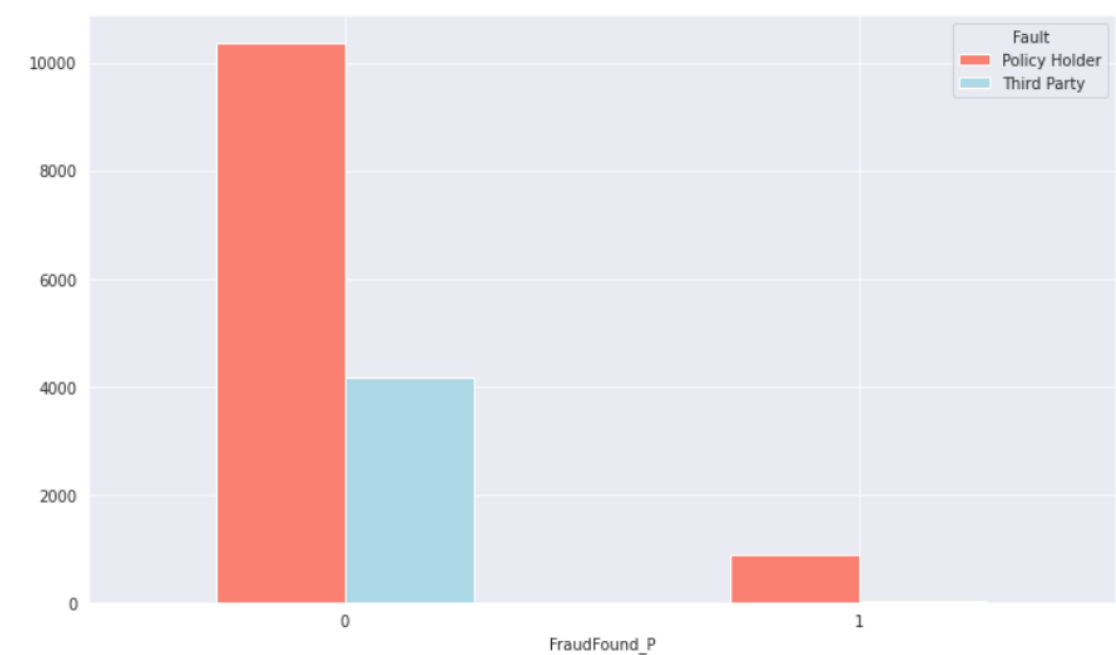
A comparison of type of insurance policy across fraudulent and non-fraudulent claims will be performed using boxplots and summary statistics

Fraud Frequency for Gender (Male vs Female)

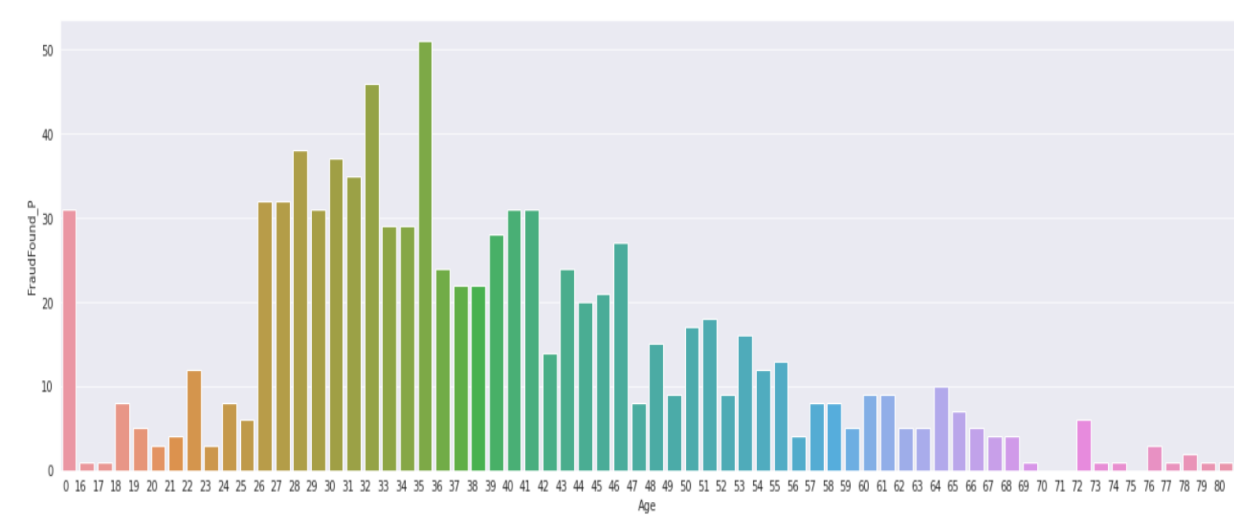
Sex	Female	Male
FraudFound_P		
0	2315	12182
1	105	818



Policy Holder vs Third Party at Fault in Insurance Fraud



Fraud Found by Age



Data Structure and Proposed Pipeline

Data Structure Classification

The dataset is classified as structured data because:

- Data is stored in rows and columns.
- Variables follow predefined formats.
- Each record represents a single insurance claim.
- Relationships among attributes are clearly defined.

The dataset does not contain:

- Unstructured text
- Audio data
- Video data
- Image data

Consequently, traditional machine learning algorithms such as Logistic Regression and XGBoost are appropriate analytical techniques.

Data Integration Plan

The project follows a multi-stage data processing pipeline:

Stage 1: Data Acquisition

Raw insurance claims data will be imported from source csv file into the python notebook.

Stage 2: Data Validation

Dataset structure, variable types, and completeness are verified.

Validation activities include:

- Duplicate detection
- Missing value analysis
- Data type verification
- Range validation

Stage 3: Data Cleaning

Cleaning procedures include:

- Handling missing values
- Correcting inconsistent category labels
- Removing duplicate records
- Standardizing variable formats

Stage 4: Feature Engineering

New analytical variables may be created, including:

- Claim-to-premium ratios
- Vehicle age categories
- Incident frequency indicators

Stage 5: Modeling Dataset Creation

Processed features are transformed into a format suitable for machine learning models.

Constraints and Limitations

Several limitations were identified during the data audit process.

Class Imbalance

The dataset contains significantly fewer fraudulent claims than legitimate claims.

Potential impacts include:

- Biased model predictions
- Reduced fraud detection sensitivity
- Misleading accuracy scores

To address this limitation, alternative evaluation metrics such as Precision, Recall, F1-Score, and ROC-AUC will be prioritized

Missing Data

Some variables contain missing observations.

Potential impacts include:

- Reduced statistical power
- Biased estimates
- Incomplete customer profiles

Appropriate imputation techniques will be evaluated depending on the extent and pattern of missingness.

Limited Fraud Labels

Fraud labels typically reflect confirmed cases rather than all actual fraud occurrences.

Potential impacts include:

- Undetected fraud cases within the non-fraud class
- Label uncertainty
- Reduced model generalizability

This limitation is common in fraud analytics and will be considered when interpreting results.

Sampling Bias

The dataset may represent a specific insurer, region, or time period.

Consequently, findings may not generalize to all insurance products or geographic markets.

Lack of Behavioral Data

The dataset primarily contains transactional claim information and does not include:

- Social network relationships
- Historical behavioral patterns
- External fraud intelligence sources

As a result, the analysis focuses on claim-level indicators rather than organized fraud networks.

Privacy and Ethics Audit

A privacy assessment was conducted to evaluate potential risks associated with the dataset.

Personally Identifiable Information (PII) Assessment

The dataset was examined for the presence of:

- Names
- Email addresses
- Phone numbers

- Social Insurance Numbers
- Physical addresses
- Banking information

No direct personally identifiable information was identified within the dataset. Most attributes consist of aggregated policy, claim, and vehicle information.

If any direct identifiers are present, they will be removed before analysis.

Ethical Considerations

Insurance fraud detection models may generate false positives that incorrectly classify legitimate claims as fraudulent.

Potential consequences include:

- Delayed claim processing
- Unnecessary investigations
- Customer dissatisfaction

Therefore, model outputs should be treated as decision-support tools rather than automated decision-making systems. Final fraud determinations should remain subject to human review and investigation.

Architecture and Logic

Proposed Technical Pipeline:

1. Data Ingestion:

The project begins with importing the insurance claims dataset from Kaggle into a structured analytical environment (Python) where Initial data profiling will be performed to understand schema, data types, missing values, and distribution of key variables such as claim amount, incident details, and fraud labels.

2. Data Cleaning & Preprocessing:

This stage involves handling missing values, encoding categorical variables, detecting outliers, and normalizing numerical features where necessary. Feature engineering may also be applied to derive meaningful variables that improve fraud detection performance.

3. Exploratory Data Analysis (EDA):

EDA will be conducted to identify trends, correlations, and patterns associated with fraudulent and non-fraudulent claims. Visualization techniques such as heatmaps, histograms, and boxplots will be used to uncover insights and guide feature selection.

4. Machine Learning Model Development:

Multiple classification algorithms such as Logistic Regression, Decision Trees, Random Forest, and XGBoost will be implemented to predict fraudulent claims. Models will be trained on historical data and optimized using hyperparameter tuning techniques.

5. Model Evaluation & Validation:

Model performance will be assessed using evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Special emphasis will be placed on recall to minimize missed fraud cases.

6. Insights & Visualization:

Key findings will be translated into business insights using dashboards and visualizations (Power BI). Feature importance analysis will help explain model decisions and highlight fraud risk drivers.

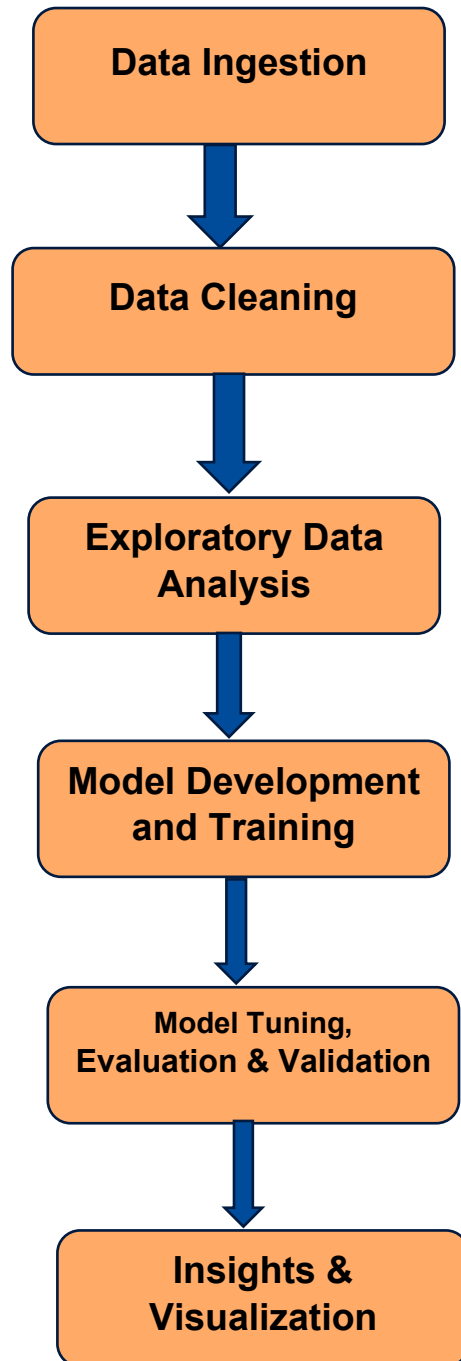
All stages are interconnected in an end-to-end workflow where insights from EDA inform feature engineering, processed data feeds machine learning models, and model outputs are translated into actionable business intelligence.

The proposed pipeline relies on a single structured dataset, widely available Python libraries, and standard machine learning workflows. Each stage is modular, allowing incremental progress and parallel work on analysis, modeling, and reporting to ensure timely completion of the capstone project.

GitHub Repository

<https://github.com/shahdata/CIND-820>

Pipeline Diagram



GenAI Declaration:

Artificial Intelligence tools, such as ChatGPT, may be used in this capstone project strictly as a supportive tool to enhance learning and productivity. AI tools will be used for purposes such as improving clarity of written content, assisting with structuring the report, generating alternative wording for technical explanations, and supporting brainstorming of analytical approaches.

However, all analytical work, including data preprocessing, exploratory data analysis, feature engineering, model development, evaluation, and interpretation of results, will be independently conducted. AI tools will not be used to generate final analytical results, conclusions, or model outputs without validation.

All content generated with AI assistance will be carefully reviewed, edited, and adapted to ensure accuracy, relevance, and alignment with academic requirements.

References

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